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💡 Thought for 1m 14s

Video Title: Reinforcement Learning Trading Bot in Python | Train an AI Agent on Forex (EURUSD)

Channel: CodeTrading

Upload Date: December 12, 2025

Views (as of retrieval): ~191,600

Duration: Approximately 20–25 minutes (based on transcript progression to ~19:53 and typical pace of code walkthroughs)

GitHub Repo (free code): https://github.com/ZiadFrancis/ReinforcementTrading_Part_1 (contains `indicators.py`, `trading_env.py`, `train_agent.py`, `test_agent.py`, `Requirements.txt`, and a `data/` folder with EUR/USD CSVs)

This is a hands-on, educational tutorial (explicitly **not financial advice**) that walks viewers step-by-step through building a complete, model-free reinforcement learning (RL) trading agent for EUR/USD for hourly data. The creator uses a gym-style environment, Stable-Baselines3's PPO algorithm, and realistic trading mechanics (spread, slippage, position persistence, SL/TP hits checked on bar high/low). The video emphasizes trial-and-error learning similar to how a human trader improves, with relative technical features to aid generalization.

1. High-Level Overview & RL Concepts (0:00 – ~3:00)

The video opens with a quick demo of the final agent: it reads historical hourly EUR/USD candles, determines long/short/hold/close actions, and learns a profitable policy through rewards based on realized P&L.

Core RL concepts explained visually and with analogies:

- **Agent:** The neural network (PPO policy) that makes trading decisions.
- **Environment:** The custom `ForexTradingEnv` that feeds observations and applies actions to historical price data.
- **Actions:** Discrete set including HOLD (0), CLOSE (1), and many OPEN variants (direction + chosen SL/TP distance).
- **Rewards:** Primarily realized net pips (after spread/commission/slippage); optional shaping from unrealized P&L while holding. Positive for winning trades, negative/zero for losses.
- **Policy:** The learned mapping from market state (window of features + position info) to action probabilities. The agent maximizes cumulative reward via trial-and-error.
- **Model-free approach:** No explicit model of the market; the agent learns directly from experience (training a dog with treats).

The host stresses this mirrors human learning: experiment, win/lose, adjust strategy. Data split: ~3 years training (2020–2023 hourly), out-of-sample testing (2023–2025). Code is split into four clean `.py` files (no Jupyter notebook) for readability.

2. Data Preparation – `indicators.py` (~3:00 – ~5:30)

The first file loads a CSV (Time (EET), OHLCV) and adds **relative, scale-invariant technical features** only (no raw price levels, to help the agent generalize across instruments/time):

- RSI(14)
- ATR(14)
- SMA(20), SMA(50)
- Slopes: `ma_20_slope = ma_20.diff()` , same for `ma_50`
- Distance from MAs: `close - ma_20` , `close - ma_50`
- MA spread and its slope

NaNs dropped; returns DataFrame + `feature_cols` list for the agent.

The host notes this is the customization point: add any indicators you want here.

3. Trading Environment – `trading_env.py` (~5:30 – ~12:00)

The most technical section. The class inherits from `gym.Env` (or `gymnasium`).

Key design choices shown and explained:

- Observation: Rolling window (default 30 bars) of the relative features + 3 internal state variable (current position $\in \{-1, 0, 1\}$, time_in_trade normalized, unrealized_pips scaled). Shape: (window_size, num_features).
- Actions (Discrete):
 - 0 = HOLD (do nothing)
 - 1 = CLOSE (if in position)
 - 2+ = OPEN (long or short + specific SL/TP from provided option lists, e.g., [60,90,120] pips video demo; code repo uses finer [5–120] grids).
- Position is **persistent** until agent closes or SL/TP is hit on the *next bar's* High/Low (conservative wins if both hit).
- Realistic costs: spread (1 pip), optional commission/slippage on entry, round-trip friction.
- Reward on close: net pips $\times$ USD-per-pip (for 1 lot $\approx$ \$10 per pip on EURUSD).
- Optional shaping: small reward/penalty for unrealized delta, hold time, open penalty.
- Random episode start + min/max episode length during training to prevent memorization.
- Equity curve tracking, trade logging.

The host runs small code snippets live to illustrate state transitions and reward calculation.

4. Training Script – `train_agent.py` (~12:00 – ~18:00)

- Loads full hourly data (repo uses 2015–2025; video shows 2020–2025 subset).
- 80/20 split (in-sample vs OOS).
- Creates vectorized envs (DummyVecEnv) with different params for train (random starts) vs eval (fixed start).
- Instantiates **PPO** (`MlpPolicy` from Stable-Baselines3).
- Trains for ~600k timesteps (video mentions ~50k in one run; checkpoints every 50k).
- Selects “best” checkpoint by final OOS equity.
- Evaluates and plots **equity curves**:
 - In-sample: steadily increasing (clear learning).
 - Out-of-sample: “acceptable” growth but flatter/more variable (realistic overfitting warning).

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on train data).

5. Testing / Inference – test_agent.py (~18:00 – end)

Simple script to load the saved best model (model_eurusd_best.zip), run one deterministic episode on test data, log closed trades to CSV, and plot the equity curve. Emphasizes using the model on fresh data.

6. Results, Discussion & Improvements (final minutes)

- **Training success:** Agent learns to increase equity via trial-and-error; reward system works.
- **Test performance:** Positive but not as strong — highlights generalization challenge (common in trading).
- **Suggested enhancements** (explicitly listed):
 - More/custom indicators in indicators.py .
 - Finer or dynamic SL/TP grids.
 - Reduce timesteps or add early stopping to fight overfitting.
 - Sliding-window cross-validation.
 - Experiment with reward shaping, different PPO hyperparameters, or other algorithms.
 - Add more friction (slippage, variable spread).

The video ends with encouragement to download the code, experiment, and combine with the creator's book/courses. Pinned resources appear on-screen.

Overall Assessment & Depth

This is a **high-quality, production-ready starter template** rather than a toy example. The environment is unusually realistic for a tutorial (bar-by-bar SL/TP checks, costs, position persistence, random start features). PPO is chosen for its stability in continuous-ish action spaces with discrete market features.

Strengths: Clean modular code, excellent explanations bridging trading intuition and RL theory, live results visualization, emphasis on out-of-sample testing and caveats.

Limitations noted in video itself: No walk-forward optimization shown, simple MLP policy, fixed data/features, potential overfitting (equity curve on test is “acceptable” not spectacular).

The project is fully reproducible (just pip install -r Requirements.txt , place your EURUSD CSV data/ , run the scripts). It serves as an outstanding practical introduction to applying RL to trading while avoiding common pitfalls like look-ahead bias or absolute price features.

If you download and run the repo, you'll get exactly what the video demonstrates — plus the ability to tweak SL/TP options, add indicators, or extend to other assets. Highly recommended for anyone interested in algorithmic trading + modern RL.

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Quick Answer X

TheRocketMan @yeobuzzfx

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